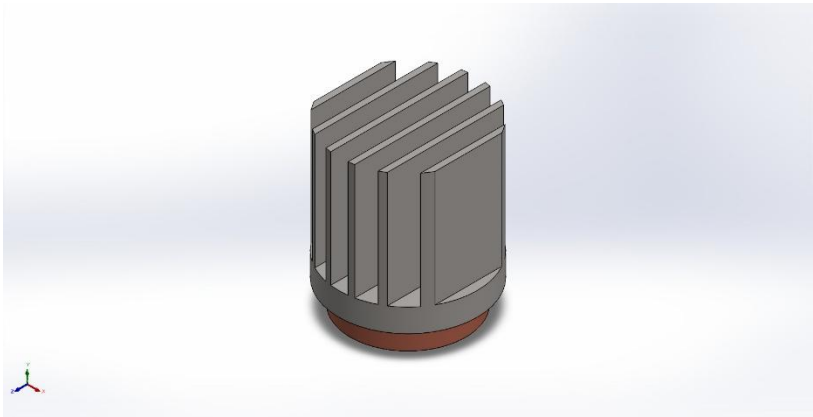




Simulation of Heat Sink

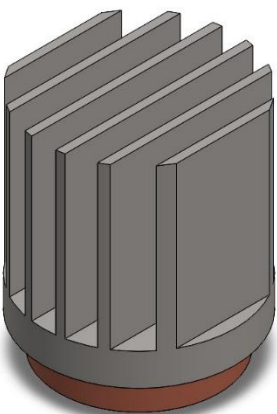
Date: 14 August 2025
Designer: Solidworks
Study name: Thermal Analysis
Analysis type: Thermal(Steady state)

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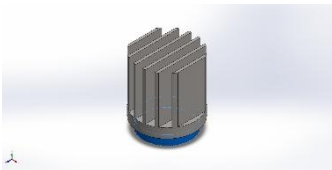
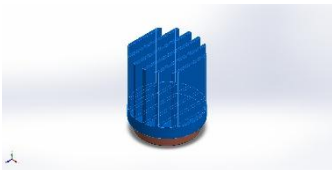


Model Information



Model name: Assem1
Current Configuration: Default

Solid Bodies

Document Name and Reference	Treated As	Volumetric Properties	Document Path/Date Modified
Boss-Extrude1 	Solid Body	Mass:1.39801 kg Volume:0.00015708 m^3 Density:8,900 kg/m^3 Weight:13.7005 N	D:\SolidWorks\FEM\Heat Sink\Part1.SLDPRT Aug 14 18:27:10 2025
Boss-Extrude2 	Solid Body	Mass:4.0129 kg Volume:0.000514474 m^3 Density:7,800 kg/m^3 Weight:39.3264 N	D:\SolidWorks\FEM\Heat Sink\Part2.SLDPRT Aug 14 18:27:11 2025



Study Properties

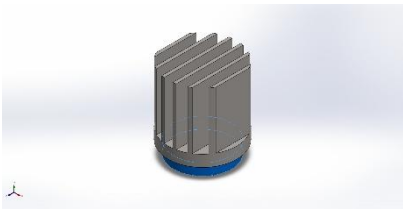
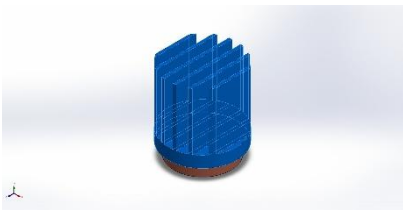
Study name	Thermal Analysis
Analysis type	Thermal(Steady state)
Mesh type	Solid Mesh
Solver type	FFEPlus
Solution type	Steady state
Contact resistance defined?	Yes
Result folder	SOLIDWORKS document (D:\SolidWorks\FEM\Heat Sink)

Units

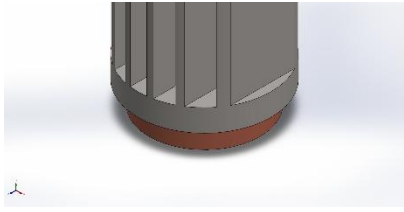
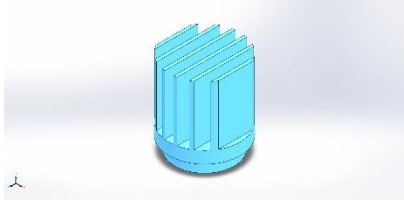
Unit system:	SI (MKS)
Length/Displacement	mm
Temperature	Kelvin
Angular velocity	Rad/sec
Pressure/Stress	N/m ²



Material Properties

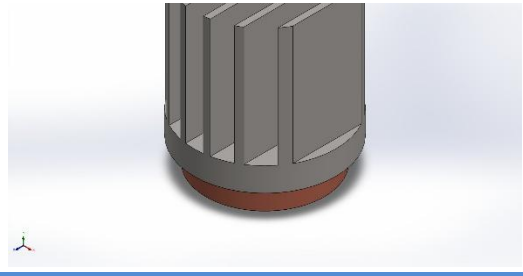
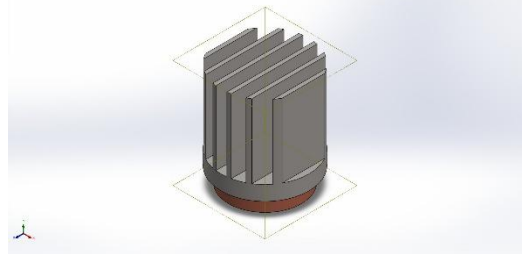
Model Reference	Properties	Components
	Name: Copper Model type: Linear Elastic Isotropic Default failure criterion: Max von Mises Stress Thermal conductivity: 390 W/(m.K) Specific heat: 390 J/(kg.K) Mass density: 8,900 kg/m ³	SolidBody 1(Boss-Extrude1)(Part1-1)
Curve Data:N/A		
	Name: Plain Carbon Steel Model type: Linear Elastic Isotropic Default failure criterion: Max von Mises Stress Thermal conductivity: 43 W/(m.K) Specific heat: 440 J/(kg.K) Mass density: 7,800 kg/m ³	SolidBody 1(Boss-Extrude2)(Part2-1)
Curve Data:N/A		

Thermal Loads

Load name	Load Image	Load Details
Heat Power-1		Entities: 1 face(s) Heat Power Value: 100 W
Convection-1		Entities: 28 face(s) Convection Coefficient: 15 W/(m ² .K) Time variation: Off Temperature variation: Off Bulk Ambient Temperature: 300 Kelvin Time variation: Off



Interaction Information

Interaction	Interaction Image	Interaction Properties
Local Interaction-1		Type: Thermal Resistance Entities: 2 face(s) Thermal Resistance: Total Value: 0.01K/W Advanced: Surface to surface
Global Interaction		Type: Bonded Components: 1 component(s) Options: Independent mesh



Mesh information

Mesh type	Solid Mesh
Mesher Used:	Blended curvature-based mesh
Jacobian points for High quality mesh	16 Points
Maximum element size	8.75923 mm
Minimum element size	8.75923 mm
Mesh Quality	High
Remesh failed parts independently	Off
Reuse mesh for identical parts in an assembly (Blended curvature-based mesher only)	Off

Mesh information - Details

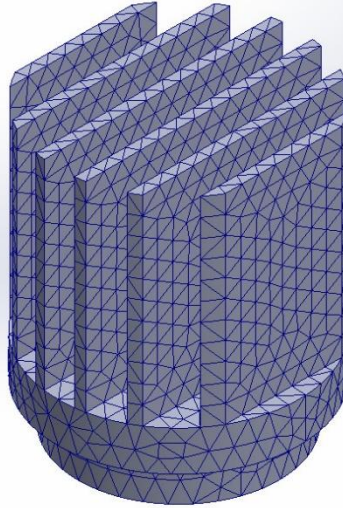
Total Nodes	19642
Total Elements	10643
Maximum Aspect Ratio	4.4362
% of elements with Aspect Ratio < 3	99.9
Percentage of elements with Aspect Ratio > 10	0
Percentage of distorted elements	0
Time to complete mesh(hh:mm:ss):	00:00:04
Computer name:	MSI

Mesh Quality Plots

Name	Type	Min	Max
Quality1	Mesh	-	-



Model name: Assem1
Study name: Thermal Analysis(-Default-)
Plot type: Mesh Quality1



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Assem1-Thermal Analysis-Quality-Quality1



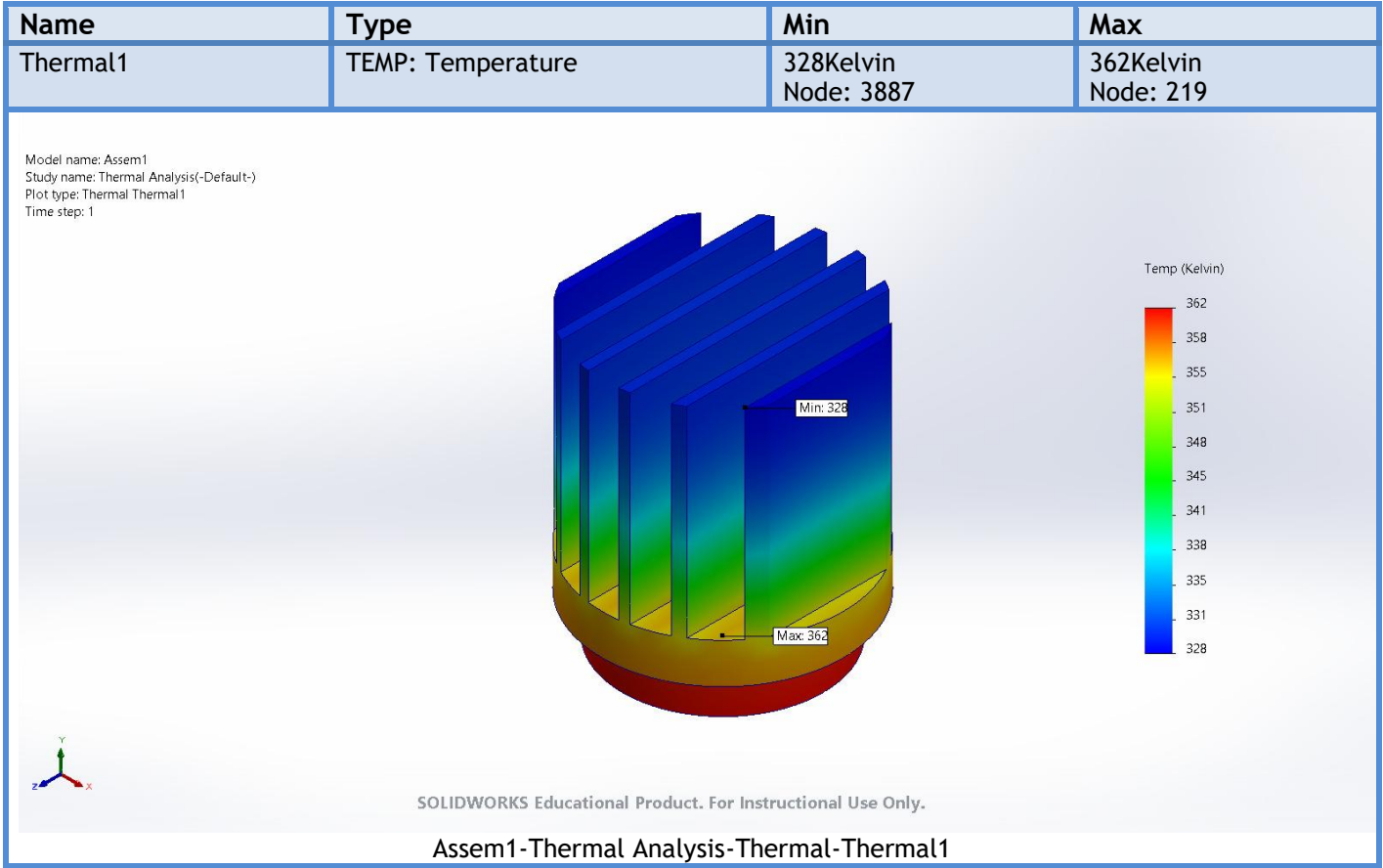
SOLIDWORKS

Analyzed with SOLIDWORKS Simulation

Simulation of Assem1

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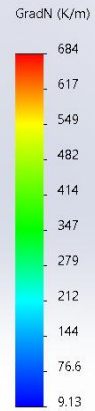
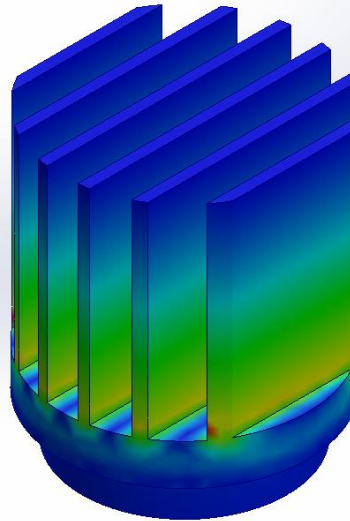
Study Results



Name	Type	Min	Max
Thermal2	GRADN: Resultant Temp Gradient	9.13K/m Node: 4062	684K/m Node: 19294



Model name: Assem1
Study name: Thermal Analysis(-Default-)
Plot type: Thermal Thermal2
Time step: 1

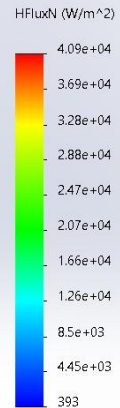
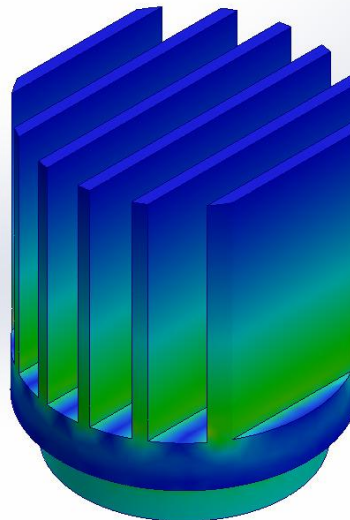


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Assem1-Thermal Analysis-Thermal-Thermal2

Name	Type	Min	Max
Thermal3	HFLUXN: Resultant heat flux	393W/m ² Node: 4062	4.09e+04W/m ² Node: 3820

Model name: Assem1
Study name: Thermal Analysis(-Default-)
Plot type: Thermal Thermal3
Time step: 1



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Assem1-Thermal Analysis-Thermal-Thermal3



